

INDU 211: TERM PAPER GROUP PRESENTATION GUIDE & 15-MINUTE VIDEO SCRIPT

"The Future of Industrial Engineering in the Generative AI Era"

Department of Mechanical, Industrial and Aerospace Engineering · Concordia University

Course: INDU 211 / Section 2X — Introduction to Production and Manufacturing Systems

Due Date: Thursday, December 1, 2026 (on Moodle)

Format: 15-Minute Recorded Video Presentation (Zoom / Microsoft Teams / OBS Studio)

1. Video Recording Guidelines & Team Plan

Parameter	Official Specification
Target Duration	Exactly 15 minutes (between 14:30 and 15:30)
Team Participation	Every team member must present an equal share
Platform	Zoom or Microsoft Teams (screen share + webcams)
Format	MP4 video recording submitted on Moodle
Style	Clear, natural speaking; do not read slides verbatim

Team Speaker Schedule (5 Members · 3 Minutes Each)

- **Speaker 1 (00:00 – 03:00):** Introduction, team members, what is Industrial Engineering, and what is Generative AI?
- **Speaker 2 (03:00 – 06:00):** Difference between traditional AI and Generative AI, and how workplaces are changing.
- **Speaker 3 (06:00 – 09:00):** Which areas of IE can be automated vs. which tasks still need human oversight.
- **Speaker 4 (09:00 – 12:00):** Serious limitations of AI: hallucinations, lack of physical common sense, and legal liability.
- **Speaker 5 (12:00 – 15:00):** Why human engineers are irreplaceable, essential skills for graduates, and conclusion.

2. Slide-by-Slide Presentation Script

Slide 1: Title & Team Introductions (00:00 – 01:30)

- **On Screen:** Slide with project title, course name (INDU 211), Concordia University logo, and names of all 5 team members.
- **Speaker 1 (Team Leader):**

"Hello everyone, and welcome to our presentation. Today, our team will be presenting on 'The Future of Industrial Engineering in the Generative AI Era: Opportunities, Limitations, and the Role of Human Engineers'.

My name is [Name], and I am presenting today along with my group members: [Name 2], [Name 3], [Name 4], and [Name 5]. Together, we are Team 4 for INDU 211.

Over the next fifteen minutes, we are going to look closely at how tools like ChatGPT, machine learning, and automation are affecting modern factories and supply chains. We will explore what tasks AI can do well, where AI fails, and why human industrial engineers are still essential for the future of manufacturing."

Slide 2: What is Industrial Engineering & The Rise of AI (01:30 – 03:00)

- **On Screen:** Simple diagram showing the core goals of IE: Eliminating Waste, Improving Productivity, Ensuring Quality, and Protecting Worker Safety. Next to it, an icon representing AI and ChatGPT.
- **Speaker 1 (Team Leader):**

"To start off, let's remember what industrial engineers actually do. Unlike other engineers who focus on single parts or machines, industrial engineers look at the whole picture. We design and improve systems that bring together people, machines, materials, and information. Our primary goals are to eliminate wasted time and materials, keep costs down, ensure product quality, and make sure workers have safe, comfortable jobs.

For decades, industrial engineers used spreadsheets, stopwatches, and math formulas to schedule production and plan factory layouts. But recently, we have seen a massive rise in Artificial Intelligence and Generative AI tools like ChatGPT and Copilot.

These tools can analyze factory data in seconds, draft maintenance reports, and even write computer code.

This raises an important question: Will AI replace industrial engineers, or will it just be another tool we use? I will now hand it over to [Name 2] to explain what this technology actually is and how it is changing workplaces."

Slide 3: Traditional AI vs. Generative AI (03:00 – 04:30)

- **On Screen:** Side-by-side comparison table:
 - **Traditional AI:** Analyzes numbers, finds patterns, makes predictions (e.g., predicting when a machine breaks).
 - **Generative AI:** Creates new content, text, code, and ideas based on prompts (e.g., writing a report, drafting a schedule).
- **Speaker 2:**

"Thank you, [Name 1]. Before looking at factories, it's important to understand the difference between traditional AI and Generative AI, because people often mix them up.

Traditional AI is all about analyzing past data to predict what will happen next. A great example in manufacturing is Predictive Maintenance. We put vibration sensors on a heavy motor. The AI learns what normal vibrations feel like, and when it notices an unusual vibration, it warns the maintenance team before the motor breaks down. Another example is demand forecasting: looking at past sales to predict how many units customers will buy next month.

Generative AI, on the other hand, is much newer. Instead of just analyzing existing numbers, it generates brand new content. It can write complete computer programs, draft safety instructions, summarize long technical manuals, and answer questions written in plain English. Think of traditional AI as a super-fast calculator, and Generative AI as an assistant that can write and draft solutions for you."

Slide 4: How AI is Changing Factory Workplaces (04:30 – 06:00)

- **On Screen:** Three icons showing real workplace changes: (1) Less routine paperwork, (2) Faster data analysis, (3) Real-time troubleshooting assistants on tablets.
- **Speaker 2:**

"So how is this actually changing daily factory work?

First, it is cutting down on routine paperwork. Industrial engineers often spend hours every week writing shift logs, documenting defective parts, and updating standard operating procedures. Generative AI can take bullet points from a technician and write a clean, professional report in seconds.

Second, it makes data analysis much faster. Instead of spending an entire afternoon writing complex Excel formulas or database queries, an engineer can simply ask the AI: 'Which assembly line had the highest scrap rate this week?' and get an answer with a graph immediately.

Third, technicians on the factory floor can use AI on their tablets to troubleshoot machines. If a machine gives an error code, the AI can immediately pull up the right page of the repair manual and show step-by-step photos.

Now, [Name 3] will explain which areas of industrial engineering can be automated by these tools."

Slide 5: What Areas of Industrial Engineering Can Be Automated? (06:00 – 07:30)

- **On Screen:** Summary table showing:
 - **High Automation:** Production scheduling, inventory reordering.
 - **Medium Automation:** Quality inspection, facility layout design, line balancing.
 - **Low Automation:** Ergonomics, worker safety, continuous improvement (Kaizen).
- **Speaker 3:**

"Thanks, [Name 2]. As industrial engineers, we study many different subjects. Some of these tasks can be almost entirely automated by AI, while others still strongly require human judgment.

Tasks that involve pure math and number crunching have the highest potential for automation. For example, production scheduling—deciding which order goes on which machine to minimize setup time—is a math puzzle that algorithms solve in seconds. Similarly, inventory reordering can easily be automated by algorithms that track warehouse stock levels and send orders to suppliers automatically.

Other areas, like quality inspection, are partially automated. Computer vision cameras can inspect hundreds of parts a second on a conveyor belt to check for surface scratches or missing labels. But when a problem keeps repeating, a computer cannot fix the root cause—a human engineer has to go investigate the machine."

Slide 6: Tasks That Still Need Humans (07:30 – 09:00)

- **On Screen:** Photos of a real factory floor: an engineer talking to a worker at an assembly station, and a layout drawing showing physical obstacles like building columns and fire exits.

- **Speaker 3:**

"On the other hand, areas like facility layout, line balancing, and ergonomics still require human engineers.

For example, an AI program can generate fifty different floor plans for a factory to reduce forklift driving distances. But the AI does not know that there is a support column right where it placed a heavy machine, or that the floor ceiling is too low for an overhead crane. An engineer has to walk the floor and check the physical realities.

The same is true for ergonomics and worker safety. AI can watch a video camera and notice if someone is bending their back improperly. But AI cannot feel what it's like to stand on a concrete floor for eight hours, or understand why a worker finds a hand tool uncomfortable. Listening to workers, redesigning tools, and keeping people safe is fundamentally human work.

Next, [Name 4] will discuss the serious errors and risks that come with AI tools."

Slide 7: AI Hallucinations and Lack of Common Sense (09:00 – 10:30)

- **On Screen:** Two warning signs:
 1. **Hallucination:** AI making up incorrect numbers while sounding confident.
 2. **No Common Sense:** AI scheduling all maintenance at once without enough technicians.

- **Speaker 4:**

"Thank you, [Name 3]. While AI is impressive, it has several dangerous limitations that every engineer must know about.

The biggest technical issue is called 'hallucination'. Generative AI models work by predicting words that sound natural, not by verifying facts in a database. That means an AI can state completely wrong numbers or fake technical specs with complete confidence.

For example, if you ask an AI model for the temperature limit of a specific plastic gear, it might give you an exact number like 200°C. If that plastic actually melts at 120°C, and the engineer does not verify it in the manufacturer's catalog, the machine could fail or catch fire.

Another major limitation is that AI has no physical common sense. It lives in computer code, so it doesn't know that two machines cannot occupy the same physical space, or that a maintenance team with two workers cannot fix five broken machines at the same time."

Slide 8: Data Privacy, Black Boxes, and Legal Liability (10:30 – 12:00)

- **On Screen:** Graphic highlighting:
 - Proprietary data security (leaking trade secrets to public clouds).
 - The "Black Box" problem (no clear step-by-step proof for regulators).
 - Legal responsibility (only a human engineer holds an engineering license and carries liability).

- **Speaker 4:**

"There are also serious legal and privacy risks.

First, data privacy: Manufacturing companies have trade secrets, like secret product formulas and proprietary machine designs. If an employee pastes company data into a free online AI tool, that information could be stored on public servers and leaked to competitors.

Second is the 'Black Box' problem. Modern deep learning models give an answer, but they cannot explain step-by-step why they made that decision. In regulated industries like aerospace or pharmaceuticals, government inspectors require full mathematical proof that a process is safe. You cannot tell an inspector that 'the computer told us to do it'.

Finally, legal liability. If an AI creates a bad schedule or layout and a worker gets hurt, you cannot sue the software. In Canada, only a licensed Professional Engineer (P.Eng.) has the legal responsibility to sign off on engineering plans. The human engineer carries all the responsibility.

Now, [Name 5] will explain the role of human engineers and the skills students need."

Slide 9: Why Human Engineers Are Irreplaceable (12:00 – 13:30)

- **On Screen:** Diagram showing "Human-in-the-Loop": AI does the number-crunching \$to\$ Human engineer verifies and makes final decisions \$to\$ Factory runs safely and smoothly.
- **Speaker 5:**

"Thank you, [Name 4]. Given all these points, what is the true role of the human engineer?

The answer is that the engineer's role becomes more important than ever, but our focus shifts. In modern manufacturing, we use what is called the 'Human-in-the-Loop' model. The AI acts as a smart assistant: it crunches massive datasets, spots patterns, and suggests a few possible options. The human engineer acts as the supervisor and final judge: checking for common sense, verifying safety, and choosing the best solution.

Human engineers are also needed for real-world emergencies. When an unexpected crisis happens—like a sudden parts shortage or a severe snowstorm—AI algorithms fail because they have no historical data to follow. Humans excel at creative problem-solving and talking to suppliers to keep production moving.

Finally, engineering is about teamwork and people. You cannot motivate a team of factory workers or negotiate with suppliers using an algorithm. Empathy, leadership, and communication are uniquely human strengths."

Slide 10: Essential Skills for Students & Conclusion (13:30 – 15:00)

- **On Screen:** List of top skills:
 - Basic Python & Data Skills
 - Classical IE Math (Operations Research, Quality Statistics)
 - Critical Thinking & Auditing AI outputs
 - Communication and Teamwork
- **Speaker 5:**

"To wrap up, what should students like us do to prepare for the future?

First, learn basic programming and data literacy. Python is becoming as common as Excel. Being able to write simple scripts to analyze factory data will give you a big advantage.

Second, master your core engineering math. The better you understand operations research, statistics, and quality control, the easier it will be to spot when an AI tool is making a mistake.

Third, develop strong communication and leadership skills. Being able to explain technical ideas clearly to managers and machine operators is something AI will never replace.

In conclusion, Artificial Intelligence and Generative AI are powerful tools that will change how factories operate. But AI is just a tool—it cannot replace human common sense, ethics, and leadership. As the saying goes: 'AI will not replace industrial engineers, but industrial engineers who know how to use AI will replace those who do not.'

Thank you very much for your time. We are now happy to take any questions."